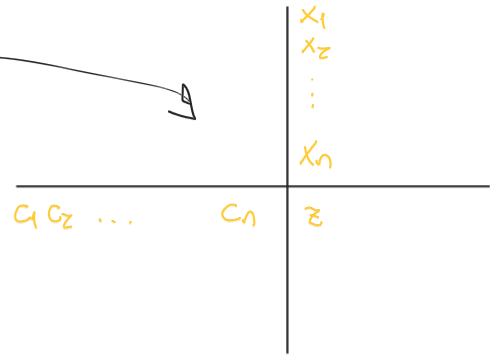
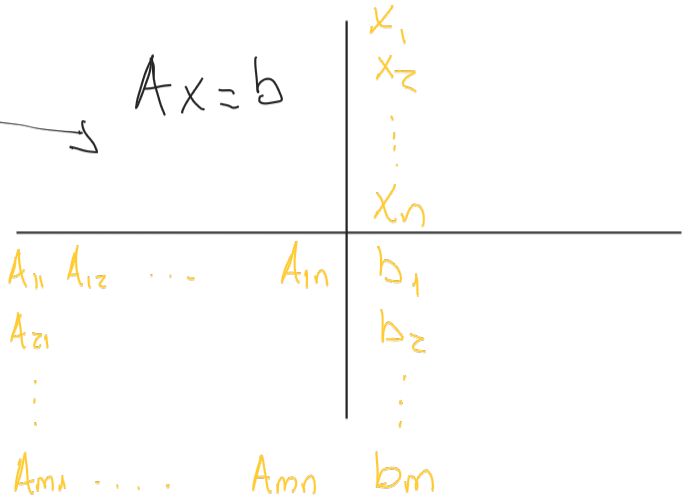
FACTIBILIDAD CAP 3.

ESTÁNDAR: MAX $z = cx$ (ESC; VEC, VEC)

S.a $Ax = b$ (MATRIZ, VEC)

$x \geq 0$





n variables

m restricciones

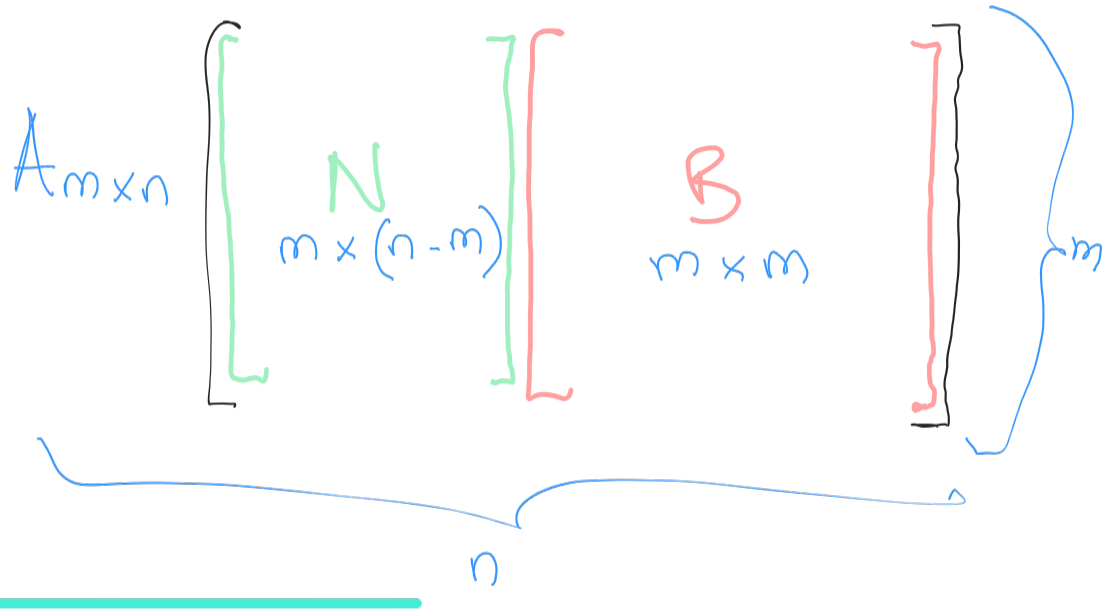
$n > m \Rightarrow$ 

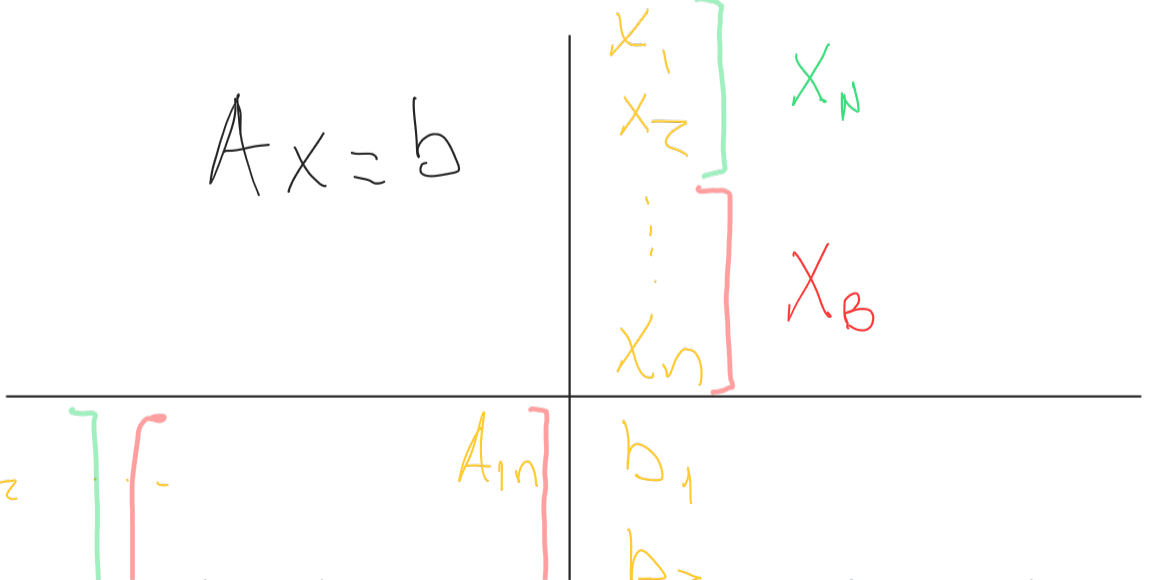
LIBRO CAP 3: A_i columnas de B

A_j columnas de N

$x_N = 0$

$x_B > 0$





A_{m1} ... A_{mn} | b_m

$Ax = b$

$B x_B + N x_N = b$, $x_N = 0$

(2) $B x_B = b$

$N = BY$

Y_{11} ... Y_{1j} ... $Y_{1(n-m)}$

Y_j

Y_{m1} ... Y_{mj} ... $Y_{m(n-m)}$

B_{11} B_{12} ... B_{1m} | N_{11} ... N_{1j} ... $N_{1(n-m)}$

N_{21}

N_j

B_{m1} ... B_{mm} | N_{m1} ... N_{mj} ... $N_{m(n-m)}$

$N_j = B Y_j$ (1)

$B^{-1} N_j = B^{-1} B Y_j$

$B^{-1} N_j = Y_j$

---

(1) $BY_j = N_j$

(2) $B x_B = b$

$(2) - (1)\cdot\theta_j$ (ESCALAR)

$B x_B - B Y_j \theta_j = b - N_j \theta_j$

$B(x_B - Y_j \theta_j) + N_j \theta_j = b$

MATRIZ, VEC, VEC, ESC, VEC, ESC, VEC

θ es escalar ; $\theta > 0$

θ_j está ASOCIADO A LA columna j

$x_1 - Y_{1j}\theta_j$

$x_2 - Y_{2j}\theta_j$

$x_m - Y_{mj}\theta_j$

B_{11} B_{12} ... B_{1m}

B_{21}

B_{m1} ... B_{mm}

$$B\left(X_B - Y_j \theta_j\right) + N_j \theta_j = b$$

$$B_1\left(X_1 - Y_{1j}\theta_j\right) + B_2\left(X_2 - Y_{2j}\theta_j\right) + \ldots + B_r\left(X_r - Y_{rj}\theta_j\right) + \ldots + B_m\left(X_m - Y_{mj}\theta_j\right) + N_j\theta_j = b$$

B_1 ↑ COLUMNA B_1

$$\theta_j = \min\left\{\frac{X_i}{Y_{ij}},\ Y_{ij} > 0;\ i = 1 \ldots m\right\}$$

CONDICIÓN DE FACTIBILIDAD

Supongamos $\theta_j = \dfrac{X_r}{Y_{rj}}$

$= X_r - Y_{rj}\theta_j$

$$B_1\left(X_1 - Y_{1j}\frac{X_r}{Y_{rj}}\right) + B_2\left(X_2 - Y_{2j}\frac{X_r}{Y_{rj}}\right) + \ldots + \underbrace{B_r\left(X_r - Y_{rj}\frac{X_r}{Y_{rj}}\right)}_{=0} + \ldots + B_m\left(X_m - Y_{mj}\frac{X_r}{Y_{rj}}\right) + N_j\frac{X_r}{Y_{rj}} = b$$

Problema Original

$\text{Max } z = CX$

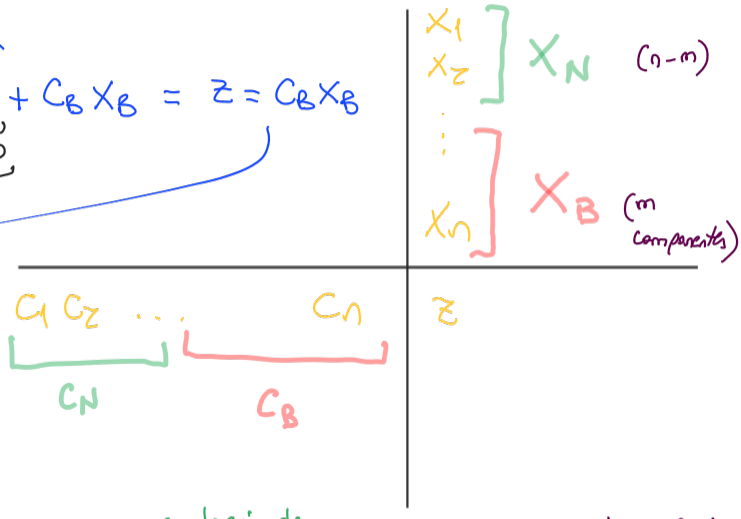
s.a $Ax = b$

$b \geq 0$

$x \geq 0$

$z = CX$

$z = C_N X_N + C_B X_B = z = C_B X_B$ ($C_N X_N = 0$, $= 0$)



$z = C_B X_B$

$z = \sum_i C_i X_i$

ver condición de factibilidad → $\sum_i B_i\left(x_i - y_{ij}\theta_j\right) + N_j\theta_j = b$ DE LA NO BASE

$\theta_j \rightarrow z' = \sum_i \left[C_i\left(x_i - y_{ij}\theta_j\right)\right] + C_j\theta_j$

$= \sum_i \left[C_i x_i - C_i y_{ij}\theta_j\right] + C_j\theta_j$

$= \sum_i C_i x_i - \theta_j \sum_i C_i y_{ij} + C_j\theta_j$

$= \ldots + \theta_j\left(C_j - \sum C_i y_{ij}\right)$

$= \sum_i c_i x_i + \theta_j \left(c_j - \underbrace{\sum_i \ldots}_{z_j} \right)$

$= \sum_i c_i x_i + \theta_j \left(c_j - z_j \right)$

$c_j - z_j > 0$

nuevo z es mejor

$\Rightarrow$ sigo iterando

$c_j - z_j \leq 0$

nuevo z no es mejor

$\Rightarrow$ NO PUEDO SEGUIR MEJORANDO

$\Rightarrow$ ESTAMOS EN EL ÓPTIMO